

# FLORENT TARIOLLE

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*Engineering student seeking a 6-month final-year research internship in deep learning, starting in 2027.*

## EDUCATION

|                                                                           |                 |
|---------------------------------------------------------------------------|-----------------|
| <b>INSA Rouen Normandy</b>                                                | Rouen, France   |
| Engineering degree, Computer Science and AI, expected 2027                | 2023 - 2027     |
| <b>University of Rouen Normandy</b>                                       | Rouen, France   |
| M.Sc. in Data Science, research-oriented MLAI track, expected 2027        | 2026 - 2027     |
| <b>Lycée Dupuy de Lôme</b>                                                | Lorient, France |
| CPGE MP (Preparatory Classes for Grandes Écoles, Mathematics and Physics) | 2021 - 2023     |

## PUBLICATIONS

**Opportunistic Target Selection** May 2026

F. Tariolle and F. Yger. *CAp 2026 (Conférence sur l'Apprentissage automatique)*.

- Introduced a lightweight wrapper for score-based black-box adversarial attacks using early class confidence signals to reduce class drift in undirected random-search attacks.
- Improved attacks by up to 27 percentage points in success rate and a 43% relative reduction in censored-mean iterations on ResNet-50, validated across 4,500 ImageNet attack runs on five standard classifiers.

## EXPERIENCE

**InterDigital** Rennes, France  
*Deep Learning Research Intern* May 2026 - Present

- Working on deep learning methods for next-generation video compression, focusing on VVC (H.266) in-loop filtering and decoder-side filter selection to improve reconstructed video quality and coding efficiency.

**Enedis** Rouen, France  
*Machine Learning Intern* Jan 2026 - May 2026

- Working on deep generative models and large-scale pipelines to simulate high-frequency time series for 38 million customers, supporting a digital twin of the French electricity grid.

## RESEARCH PROJECTS

**DashVMC: Real-Time Discrete World-Model Control in Geometry Dash** Feb 2026 - Present  
*Independent research project; manuscript in preparation.*

- Built a real-time Geometry Dash Vision-Model-Controller system with FSQ tokenization, action-conditioned transformer dynamics, and an actor-critic trained from behavioural cloning plus PPO in latent FSQ-token rollouts.
- Deployed the learned controller on live gameplay at 30 FPS; decoded pixel rollouts are used only for qualitative demos, including plausible level continuations from real gameplay prefixes.
- Studied Structured Label Smoothing as an FSQ-neighbour prior; a matched IRIS/Pong diagnostic showed partial stability signals but no robust general improvement over cross-entropy.

## AWARDS

**Finalist, Hack the World(s) Hackathon** Jun 2026  
*24-hour team project at a Yann LeCun-sponsored hackathon on JEPAs and world models*

- Developed a geometry-aware joint-embedding EEG SSL system: strong SIGReg frozen-transfer baselines and Riemannian latent analysis showing structure in SPD covariance representations.

## OPEN SOURCE

**Rose** (Active: 15K+ DAU) Sep 2025 - Present  
*Co-founder & Lead Developer*

- Co-founded and led development of a high-traffic real-time customization tool used by 15K+ daily active users; coordinated a 6-person team and designed the Python backend, Cloudflare Workers WebSocket relay, and browser-side JavaScript integration.
- Replaced OCR-based state extraction with deterministic DOM parsing, reducing latency and operational complexity.

## TECHNICAL SKILLS

**Programming:** Python, C++

**AI & ML:** PyTorch, CUDA; Deep Learning, Self-Supervised Learning, World Models, Model-Based RL and Planning

**Languages:** French (native), English (TOEIC 970/990)

## REFERENCES

**Florian Yger** florian.yger@insa-rouen.fr  
**Frédéric Lefebvre** frederic.lefebvre@interdigital.com